

What is Time Series Analysis?

Time Series Analysis is the process of analyzing data collected over time to identify patterns and predict future values.

Simple Definition

Past Data + Time



Find Pattern



Predict Future

Example: Ice Cream Sales

Month	Sales
Jan	100
Feb	120
Mar	150
Apr	180
May	220
Jun	250

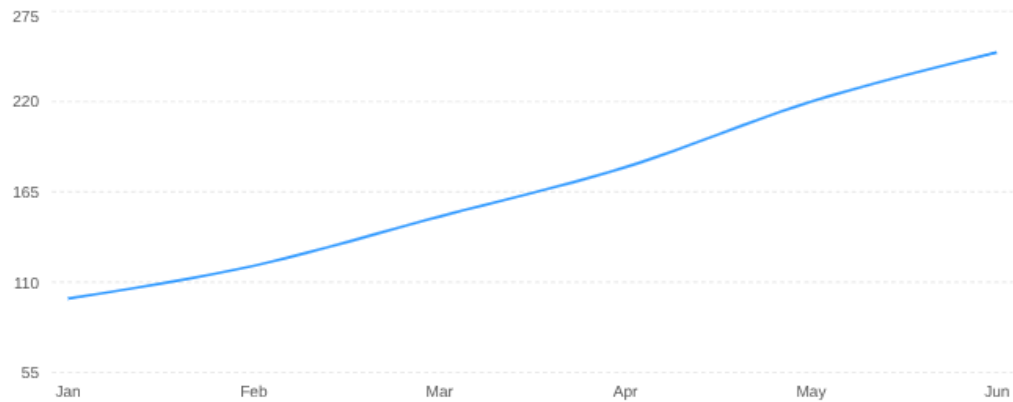
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We can see sales are increasing over time.

Trend Visualization

Ice Cream Sales Trend

Example of time series data collected over months.



Using this trend, we can estimate future sales.

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How It Works

Historical Data
↓
Analyze Pattern
↓
Identify Trend
↓
Forecast Future

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Real-Life Example

Stock Market

Day	Share Price
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Monday	₹100
Tuesday	₹105
Wednesday	₹110
Thursday	₹115

Time series models can help forecast future prices.

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Where is Time Series Analysis Used?

Domain	Example
Stock Market	Predict share prices
Weather Forecasting	Predict temperature and rainfall
Sales Forecasting	Predict future product sales
Electricity Usage	Predict power demand
Traffic Analysis	Predict traffic congestion
Healthcare	Monitor patient heart rate
Banking	Detect unusual transactions
E-commerce	Predict future customer demand
Agriculture	Forecast crop yield
Tourism	Predict tourist arrivals

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Example Applications

Weather Prediction

Past Temperature Data



Time Series Analysis



Tomorrow's Temperature

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Stock Market Prediction

Past Share Prices



Time Series Analysis



Future Share Price

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Sales Forecasting

Past Monthly Sales



Time Series Analysis



Next Month Sales

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Components of Time Series

Component	Meaning
Trend	Long-term increase or decrease
Seasonality	Repeating pattern (summer, winter, festivals)
Cyclic Pattern	Long-term business cycles
Noise	Random fluctuations

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Ice Cream Sales

Summer ↑ High Sales

Winter ↓ Low Sales

This repeating pattern is called Seasonality.

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Simple Diagram

Time Series Analysis

Past Data



Analyze Trends



Find Patterns



Forecast Future



Predicted Value

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Time Series Analysis is a technique used to analyze data collected over time and predict future values based on historical patterns and trends.

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